ESA 321
Aerospace Structures

Lecture 3 – Bar Element

Objectives

- **To apply FEM using bar element for solving structural problems.**

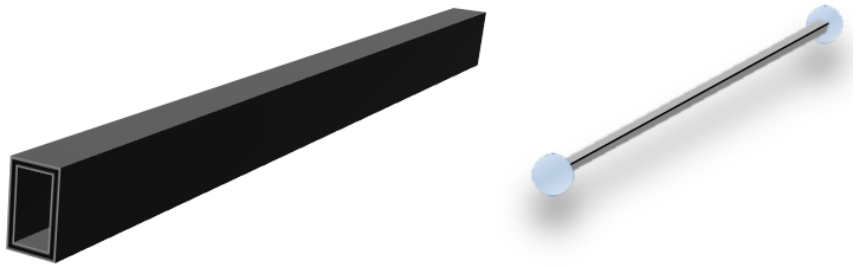


Figure 1: 1D Beam/Bar Element Representation

1D – Bar Element ~~(Truss Element)~~

Properties of a bar element

1 Deformation (DOF)
The deformation (degree of freedom) is along the **axial direction**.



2 Load application
The load can only be applied at the **two ends**.



3 Type of load
A bar can only be subjected to **axial load** (tension or compression).



4 Force transmission
Forces can only be transmitted along the **axial direction**.



5 Cannot sustain shear and moment
A bar element cannot sustain **shear forces** or **bending moments**.

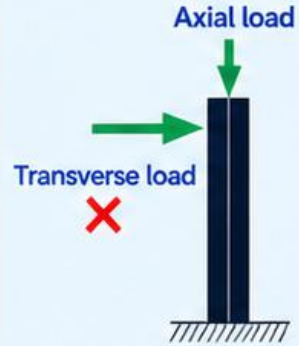


6 Mass
The mass of the bar element is neglected.



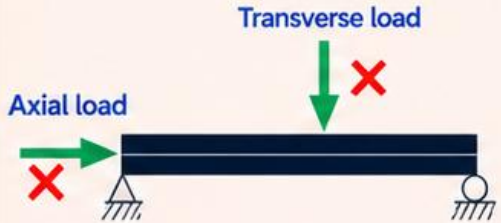
Difference between Axial Loading & Transverse Loading

Axial Loading



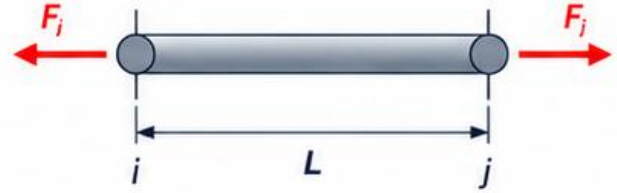
Bar element carries axial load only ✓

Transverse Loading



Bar element cannot carry transverse load, shear or bending moment ✗

Typical 1D Bar Element Model



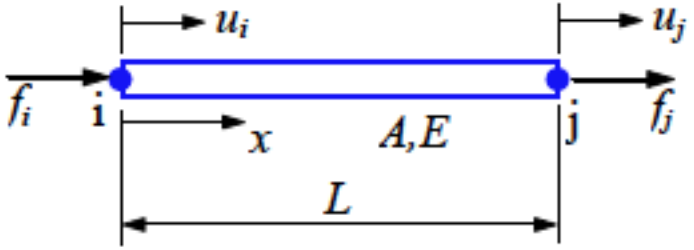
Axial stiffness

$$k = \frac{AE}{L}$$

A = cross-sectional area (m^2)
E = Young's modulus (Pa)
L = element length (m)

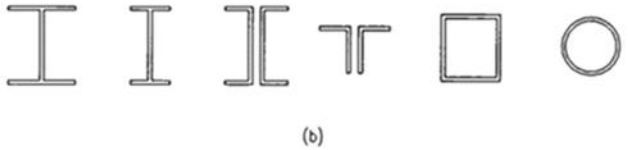
1D - Bar Element (1)

Consider a uniform prismatic bar:



L	length
A	cross-sectional area
E	elastic modulus
$u = u(x)$	displacement
$\varepsilon = \varepsilon(x)$	strain
$\sigma = \sigma(x)$	stress


(a)


(b)


(c)

Strain-displacement relation:

$$\varepsilon = \frac{du}{dx}$$

Stress-strain relation:

$$\sigma = E\varepsilon$$

1D - Bar Element (4)

- The stiffness of bar element can be derived as

$$k = \frac{EA}{L}$$

- Therefore, bar element stiffness equation is

$$\mathbf{k} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Complete formulation for one bar element is

$$\frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} u_i \\ u_j \end{Bmatrix} = \begin{Bmatrix} f_i \\ f_j \end{Bmatrix}$$

Bar element

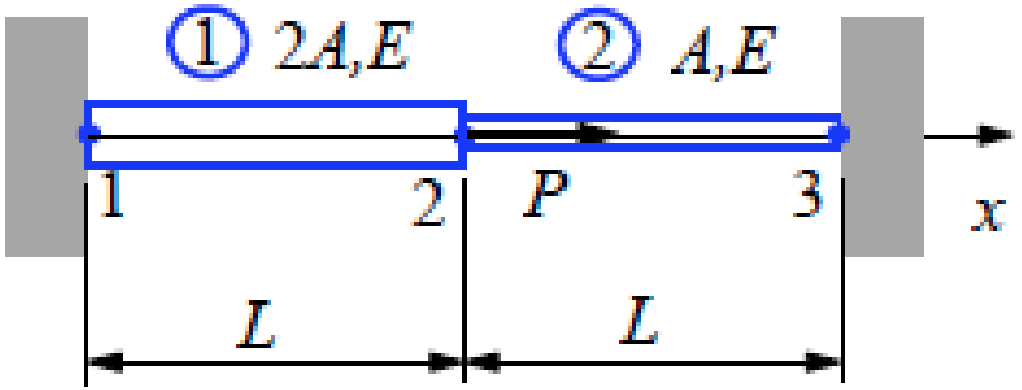
Not the same

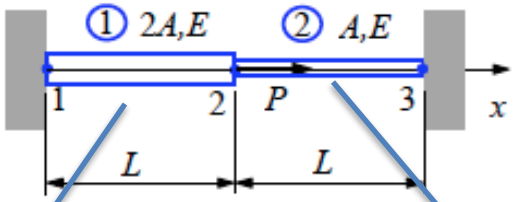
$$\begin{bmatrix} k & -k \\ -k & k \end{bmatrix}\begin{Bmatrix} u_i \\ u_j \end{Bmatrix} = \begin{Bmatrix} f_i \\ f_j \end{Bmatrix}$$

Spring Element Equation

Example

- Find the displacement of node 2 for the system shown using the **expression** of P, L, E & A using bar element discretization.





$$\frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} u_i \\ u_j \end{Bmatrix} = \begin{Bmatrix} f_i \\ f_j \end{Bmatrix}$$

Element 1

$$\mathbf{k}_1 = \frac{2EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Element 2

$$\mathbf{k}_2 = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{matrix} u_1 & u_2 \end{matrix}$$

$$\frac{EA}{L}\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix}$$

$$\begin{matrix} u_2 & u_3 \end{matrix}$$

$$\frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} f_2 \\ f_3 \end{Bmatrix}$$

$$\begin{matrix} u_1 & u_2 & u_3 \end{matrix}$$

$$\frac{EA}{L}\begin{bmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\begin{matrix} u_1 & u_2 & u_3 \end{matrix}$$

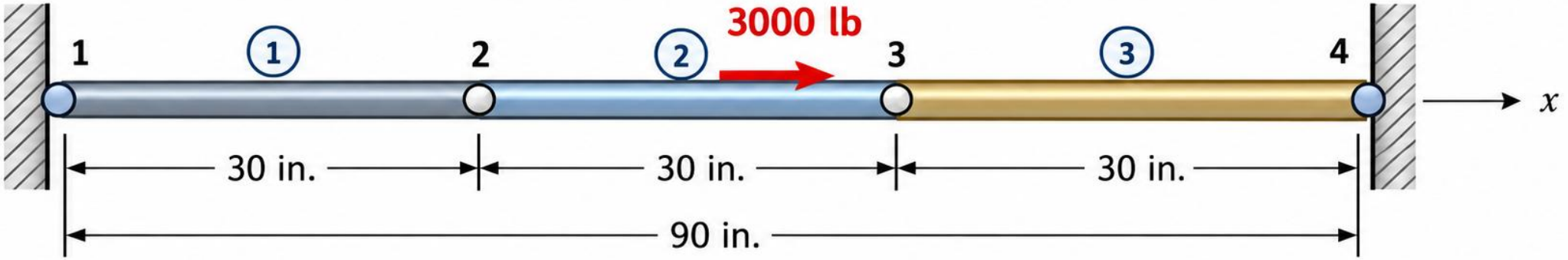
$$\frac{EA}{L}\begin{bmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\frac{EA}{L}[3]\{u_2\} = \{P\}$$

$$u_2 = \frac{PL}{3EA}$$

1D - Bar Element (Practice)

Consider the following three-bar system shown below. Assume for elements 1 and 2: $A = 1\ in^2$ and $E = 30 \times 10^6\ psi$ and for element 3: $A = 2\ in^2$ and $E = 15 \times 10^6\ psi$.



Elements 1 and 2
$A = 1\ in^2,\ \ E = 30 \times 10^6\ psi$

Element 3
$A = 2\ in^2,\ \ E = 15 \times 10^6\ psi$

Determine:

a. the global stiffness matrix,
b. the displacements of nodes 2 and 3,
c. the reactions at nodes 1 and 4.

1D - Bar Element (17)

For elements 1 and 2:

$$\begin{matrix} 1 & 2 \end{matrix} \quad \text{node numbers for element 1}$$
$$\begin{matrix} 2 & 3 \end{matrix} \quad \text{node numbers for element 2}$$

$$\mathbf{k}^{(1)} = \mathbf{k}^{(2)} = \frac{(1)\left(30\times10^{6}\right)}{30}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} {}^{lb}\!/_{in} = 10^{6}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} {}^{lb}\!/_{in}$$

For element 3:

$$\begin{matrix} 3 & 4 \end{matrix} \quad \text{node numbers for element 3}$$

$$\mathbf{k}^{(3)} = \frac{(2)\left(15\times10^{6}\right)}{30}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} {}^{lb}\!/_{in} = 10^{6}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} {}^{lb}\!/_{in}$$

As before, the numbers above the matrices indicate the displacements associated with the matrix.

1D - Bar Element (18)

Assembling the global stiffness matrix by the direct stiffness methods gives:

E1 E 2 E 3

$$\mathbf{K} = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

Relating global nodal forces related to global nodal displacements gives:

$$\begin{Bmatrix} F_{1x} \\ F_{2x} \\ F_{3x} \\ F_{4x} \end{Bmatrix} = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix}$$

1D - Bar Element (19)

The boundary conditions are: $u_1 = u_4 = 0$

$$\begin{Bmatrix} F_{1x} \\ F_{2x} \\ F_{3x} \\ F_{4x} \end{Bmatrix} = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ u_2 \\ u_3 \\ 0 \end{Bmatrix}$$

Applying the boundary conditions and the known forces ($\boldsymbol{F}_{2x}$ = 3000 *lb*) gives:

$$\begin{Bmatrix} 3000 \\ 0 \end{Bmatrix} = 10^6 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix}$$

1D - Bar Element (20)

Solving for u_2 and u_3 gives: $u_2 = 0.002\,in$

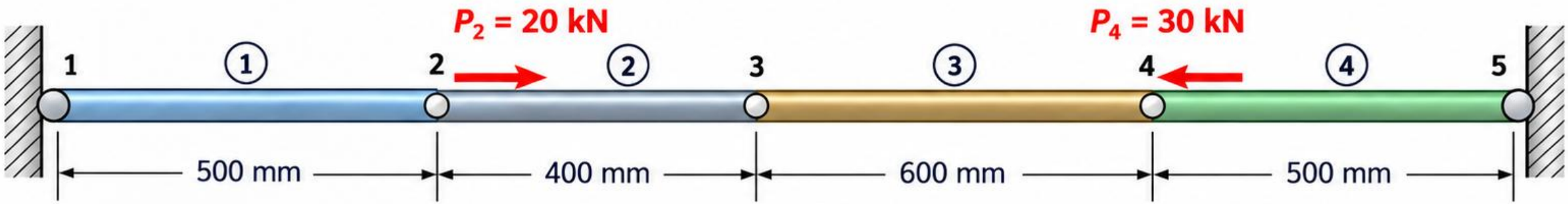
$u_3 = 0.001\,in$

The global nodal forces are calculated as:

$$\begin{Bmatrix} F_{1x} \\ F_{2x} \\ F_{3x} \\ F_{4x} \end{Bmatrix} = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ 0.002 \\ 0.001 \\ 0 \end{Bmatrix} = \begin{Bmatrix} -2000 \\ 3000 \\ 0 \\ -1000 \end{Bmatrix} lb$$

Practice Question 1
Bar Element

Consider the following four–bar axial system shown below. Both ends are fixed.



Element properties

Element	Nodes	Length, *L* (mm)	Area, *A* (mm²)	Young's modulus, *E* (GPa)
1	1 – 2	500	400	200
2	2 – 3	400	300	200
3	3 – 4	600	500	200
4	4 – 5	500	400	200

Boundary conditions

$$u_1 = 0$$
$$u_5 = 0$$

Determine:

- **a** The element stiffness matrix for each bar.
- **b** The global stiffness matrix ***K***.
- **c** The displacements u_2, u_3 and u_4.
- **d** The axial force and stress in each element.
- **e** The reaction forces at nodes 1 and 5.